Preclusters, maximal invariant polyhedral supports and siphons: exact criteria and counterexamples for the precise forms of Iosif's duality conjecture

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Abstract. For conservative mass-action networks with a positive steady state and no two species with the same rate, Iosif (arXiv:2603.08767, Conjecture 22 and Question 24) conjectured a "duality relation" between preclusters, which are sets of reactions, and maximal invariant polyhedral supports (MIPS), which are sets of species. Since species and rate constants exchange roles in his dual network, he also conjectured that preclusters and siphons are dual objects. The relation is not defined in the paper, so we make each ingredient precise. Definition 17 and Remark 18 of the paper agree: MIPS are the classes of species whose columns in a matrix of conservation laws are positively proportional. The precluster construction of the paper and the iterated one of the author's thesis give the same partition. Both hypotheses, conservativity and consistency, are exchanged by the duality. We then prove exact criteria. The preclusters of a consistent network refine the MIPS of its dual if and only if any two reactions with a common source are fully coupled. The two partitions coincide if and only if, in addition, every nontrivial class of fully coupled reactions contains such a pair. The dual statements hold for species. Consequently every formulation we could extract fails: equality, refinement in either direction and equal cardinality, on either side of the duality and under three readings of MIPS. The counterexamples are tiny: $X_1 \rightleftarrows X_2$, the 3-cycle $X_1 \to X_2 \to X_3 \to X_1$ (weakly reversible, deficiency zero, self-dual), the one-site futile cycle, and two examples from the author's thesis. The siphon formulations of Question 24 and the claim of Remark 23 (internal cycles refine preclusters) fail on the same networks. Exact computations on 15 513 small networks confirm the criteria. The conjecture as stated, with the relation left undefined, remains open, as does its motivation: a transfer of known cases of the Global Attractor Conjecture through duality. We note three constraints on such a transfer: the duality preserves the dimension of the stoichiometric subspace, it does not preserve weak reversibility, and for weakly reversible networks the preclusters of the dual network are singletons.

The question

Iosif [1] develops a duality for mass-action networks. The dual of a network exchanges the roles of species and rate constants: the dual has one species per reaction and one reaction per species. He proves that conserved quantities are dual to internal cycles (his Theorem 21) and then states, in §3 (p. 7 of arXiv v2, 15 July 2026; identical in v1 and in HAL v3):

> *"Conjecture 22. For conservative mass-action networks that do not have two species with exactly the same rates and have at least one positive steady state, there is a duality relation between the sets of preclusters and of maximal invariant polyhedral supports.*
> *Remark 23. The condition $\dot{x}_i \neq \dot{x}_j$ for $i \neq j$ is necessary in order to make sure no further clustering on the left kernel is done. We require that there is at least one positive steady state so that the internal cycles are a refinement of preclusters.*
> *We conclude with the following question:*
> *Question 24. Is there a duality relation between preclusters and siphons?"*

The abstract and the introduction (p. 2) read: "we conjecture that there is a duality relation between preclusters and maximal invariant polyhedral supports. Given the close relation between maximal invariant polyhedral supports and siphons, we also conjecture that siphons and preclusters are dual objects." The status of the statement has changed over time. The 2020 HAL version [2] states it as a theorem, without the positive-steady-state hypothesis and without proof. A 2024 talk [3] lists "MIPS ... are dual to preclusters" as a theorem of Ananiadi and Iosif. A 2025 poster abstract [4] says "we prove that preclusters are dual to maximal invariant polyhedral supports". The 2026 preprint calls it a conjecture. None of these texts defines "duality relation". Preclusters come from the author's thesis [5], Chapter 4, after the clusters of Conradi and Flockerzi [6].

What we prove. In §1 we fix the definitions and show that the ambiguities in them are harmless (Proposition 3 and Lemma 5). In §2 we prove exact criteria for each comparison between preclusters and MIPS (Theorem 7 and Corollary 8). In §3 we use them to refute every formulation listed in Table 11 (Theorem 1). §4 reports an exhaustive exact check on small networks and §5 lists what remains open. Throughout, $A \preceq B$ means that the partition (or family) A refines B: every block of A is contained in a block of B.

Theorem 1 (main result). Read "duality relation" as any of the relations of Table 11: equality, refinement in either direction, or equal number of blocks, between the preclusters of $\mathcal{N}^*$ and the MIPS of $\mathcal{N}$ (species side) or between the preclusters of $\mathcal{N}$ and the MIPS of $\mathcal{N}^*$ (reaction side). Read MIPS in any of the three ways of Remark 4. Then Conjecture 22 is false under every such reading. Question 24 has a negative answer for each of the siphon relations of Table 11, and the claim of Remark 23 is false. Every counterexample satisfies all hypotheses of Conjecture 22, has positive steady states, and has at most six species.

1 Definitions, and why their ambiguities do not matter

Networks. A network $\mathcal{N}$ has species $X_1, \dots, X_n$ and reactions $R_j : y_j \to y'_j$ ($j = 1, \dots, r$) with $y_j, y'_j \in \mathbb{Z}^n_{\geq 0}$. The educt and product matrices are $Y_{\mathrm{e}} = (y_1 | \cdots | y_r)$ and $Y_{\mathrm{p}} = (y'_1 | \cdots | y'_r)$, and $\Gamma = Y_{\mathrm{p}} - Y_{\mathrm{e}}$ is the stoichiometric matrix. The mass-action system is $\dot{x} = \Gamma \operatorname{diag}(k)\, x^{Y_{\mathrm{e}}}$ with $k \in \mathbb{R}^r_{>0}$. The *flux cone* is $C(\mathcal{N}) = \ker \Gamma \cap \mathbb{R}^r_{\geq 0}$. The *conservation cone* is $K(\mathcal{N}) = \{z \in \mathbb{R}^n_{\geq 0} : z^{\mathsf{T}}\Gamma = 0\}$. Both cones are pointed. The supports of the extreme rays of $C(\mathcal{N})$ are the internal cycles of $\mathcal{N}$ (Definition 20 of [1]; [7]). Let E be the $r \times m$ matrix whose columns are the extreme rays of $C(\mathcal{N})$, and let $n_1, \dots, n_r \in \mathbb{R}^m$ be its rows. Let W be the $p \times n$ matrix whose rows are the extreme rays of $K(\mathcal{N})$, and let $w_1, \dots, w_n \in \mathbb{R}^p$ be its columns. Positive rescaling of the rays does not affect anything below.

Hypotheses of Conjecture 22.

(H1) *Conservative*: some $z \in K(\mathcal{N})$ has all coordinates positive.

(H2) *At least one positive steady state.* The paper treats the rate constants as parameters (Definition 4 and Remark 5), so (H2) means that some k admits a positive steady state. This is equivalent to consistency: $C(\mathcal{N})$ contains a vector with all coordinates positive. Indeed, if $x > 0$ is a steady state for k, then $v = \operatorname{diag}(k)x^{Y_e} > 0$ lies in $\ker \Gamma$. Conversely, if $v > 0$ lies in $\ker \Gamma$, then $x = (1, \dots, 1)$ is a steady state for $k = v$.

(H3) *No two species with exactly the same rates*: $\dot{x}_i \neq \dot{x}_j$ for $i \neq j$, as in Remark 23. As polynomials in (k, x), $\dot{x}_i = \sum_j \Gamma_{ij} k_j x^{y_j}$, so (H3) says that the rows of Γ are pairwise distinct.

Every formulation of Table 11 is refuted by one of the networks T, W, U, D below. These satisfy the strongest reading of (H3): no row of Γ is a multiple of another, so $\dot{x}_i$ and $\dot{x}_j$ are not even proportional. Every counterexample has a positive steady state for explicit rate constants, and T, W, R and $N_{4.12}$ have one for all rate constants.

The dual network. Following §3 of [1] (see also [8]), the dual $\mathcal{N}^*$ has species $K_1, \dots, K_r$, one for each reaction of $\mathcal{N}$, and one reaction $\rho_i \colon K^{(Y_e)_{i\cdot}} \to K^{(Y_p)_{i\cdot}}$ for each species X_i of $\mathcal{N}$. Here $(Y_e)_{i\cdot}$ denotes the ith row. The dual may contain the zero complex, and it contains a loop $\rho_i \colon y \to y$ if X_i is a pure catalyst. All definitions and proofs below apply verbatim to such networks.

Lemma 2. $Y_e(\mathcal{N}^*) = Y_e^{\mathsf{T}}$, $Y_p(\mathcal{N}^*) = Y_p^{\mathsf{T}}$, $\Gamma(\mathcal{N}^*) = \Gamma^{\mathsf{T}}$ and $\mathcal{N}^{**} = \mathcal{N}$. Hence $C(\mathcal{N}^*) = K(\mathcal{N})$ and $K(\mathcal{N}^*) = C(\mathcal{N})$. In particular $\mathcal{N}$ satisfies (H1) if and only if $\mathcal{N}^*$ satisfies (H2), and vice versa. $\mathcal{N}^*$ satisfies (H3) if and only if the columns of Γ are pairwise distinct. The internal cycles of $\mathcal{N}^*$ are the supports of the extreme conservation laws of $\mathcal{N}$, which is the content of Theorem 21 of [1].

Proof. All statements follow from the definition by transposition. □

Maximal invariant polyhedral supports. Assume (H1). Let $Z \in \mathbb{R}^{q \times n}$ be a matrix whose rows form a basis of the conservation space $L = \{z : z^{\mathsf{T}}\Gamma = 0\}$. For c in the cone $Q = Z(\mathbb{R}^n_{\geq 0})$, the invariant polyhedron is $P_c = \{x \in \mathbb{R}^n_{\geq 0} : Zx = c\}$. By (H1) it is a nonempty polytope. Each vertex v of P_c is decorated by its support $\operatorname{supp} v$ (the monomial $\prod_{i \in \operatorname{supp} v} x_i$ in [1]). Distinct vertices have distinct supports: $\{Ze_t : t \in \operatorname{supp} v\}$ is linearly independent, so v is the only point of P_c with support $\operatorname{supp} v$. The *decorated type* of P_c is its face lattice with these labels. The paper's Definition 17 reads:

> *"A set of species $\Sigma = \{x_i \mid i \in I \subseteq [n]\}$ is called an invariant polyhedral support if all combinatorial types of decorated abstract invariant polyhedra are invariant under the map $x_i \mapsto x_j$ for all $x_i, x_j \in \Sigma$. An invariant polyhedral support is maximal if it is maximal with respect to inclusion."*

Remark 18 adds that "one needs to compute the chamber decomposition in the space of linear conserved quantities, and decorate each ray with the corresponding variables. The set of all variables decorating an individual ray is a maximal invariant polyhedral support." There are finitely many decorated types (the source cites [9] for this; it is

not used below). We read "invariant" as: for every $c \in Q$, relabelling the vertices of P_c by the transposition $\tau = (i\,j)$ gives the same decorated type. The next proposition shows that this reading agrees with Remark 18.

Proposition 3. Assume (H1) and let $i \neq j$. The following are equivalent.

(1) For every $c \in Q$, the decorated type of P_c is invariant under the transposition of x_i and x_j.

(2) $Ze_i = \mu Ze_j$ for some $\mu > 0$: x_i and x_j decorate the same ray of the chamber decomposition (Remark 18).

(3) $e_i - \mu e_j \in \operatorname{im}\Gamma$ for some $\mu > 0$.

(4) $w_i = \mu w_j$ for some $\mu > 0$.

(5) $z_i = \mu z_j$ for some $\mu > 0$ and every conservation law $z \in L$.

Write $i \sim j$ when these hold. Then $\sim$ is an equivalence relation. A set of species is an invariant polyhedral support if and only if it lies in one class, and the MIPS of $\mathcal{N}$ are exactly the classes. In particular the MIPS partition the species. We denote this partition by $\mathsf{M}(\mathcal{N})$.

Proof. (2)⇒(1): Let $\varphi\colon \mathbb{R}^n \to \mathbb{R}^n$ swap the coordinates i and j with weights: $\varphi(x)_i = x_j/\mu$, $\varphi(x)_j = \mu x_i$, and $\varphi(x)_t = x_t$ otherwise. Then $Z\varphi(x) = Zx$, because $\mu x_i Ze_j + (x_j/\mu)Ze_i = x_i Ze_i + x_j Ze_j$. Moreover φ preserves $\mathbb{R}^n_{\geq 0}$, and $\operatorname{supp}\varphi(x) = \tau(\operatorname{supp} x)$. So φ restricts to an affine automorphism of every P_c that relabels supports by τ, and the decorated type is invariant. (1)⇒(2): Take $c = Ze_i \in Q$. The point e_i is a vertex of P_c with label $\{i\}$. Invariance gives a vertex with label $\{j\}$, that is, a point $ae_j \in P_c$ with $a > 0$. Hence $Ze_i = aZe_j$. Note also $Ze_i \neq 0$, since $z_i > 0$ for the positive conservation law. (2)⇔(3): $\ker Z = L^\perp = \operatorname{im}\Gamma$. (2)⇔(5): the rows of Z span L. (4)⇔(5): by (H1), $K(\mathcal{N})$ contains an interior point of L, so it spans L. Being pointed, it is generated by its extreme rays, so the rows of W span L. The relation in (2) is clearly an equivalence. Definition 17 asks for invariance under every transposition of two elements of Σ, so the invariant polyhedral supports are the subsets of classes, and the maximal ones are the classes. □

Remark 4 (three readings of MIPS). Proposition 3 is the reading that matches Remark 18 ("ray reading"). Two other readings are conceivable. The *set reading* only asks that the set of all decorated types be invariant under τ. Its classes are unions of ray classes and can be strictly larger: for $Y : X_1 + 2X_2 \rightleftarrows X_3$ the ray reading gives $\{X_1\}, \{X_2\}, \{X_3\}$ and the set reading gives $\{X_1, X_2\}, \{X_3\}$. The set reading contradicts Remark 18. The *semiflow reading* takes the supports of the extreme conservation laws, that is, the internal cycles of $\mathcal{N}^*$. It is suggested by the diagram at the end of §3 of [1], which links internal cycles and MIPS by an arrow labelled "dual". For Y it gives $\{X_1, X_3\}, \{X_2, X_3\}$, which is not a partition. All three readings agree in Example 25 of [1], so that example cannot distinguish them. Our counterexamples work for all three readings (Table 11). Our scripts compute the set reading whenever $q \leq 2$. ⌟

Siphons. A *siphon* is a nonempty set S of species such that every reaction with a product species in S has a reactant species in S (Definition 13 of [1]; [10] [11]). A siphon is minimal if no proper subset is a siphon (Definition 14). We write $\mathsf{S}(\mathcal{N})$ for

the family of minimal siphons. The support of a semi-positive conservation law is a siphon [11]. This is presumably the "close relation" between MIPS and siphons mentioned in the source, at least in the semiflow reading of Remark 4.

Preclusters. Two distinct reactions with the same source form a *doubling pair* (thesis terminology [5]). Let G be the graph on $[r]$ whose edges are the doubling pairs. The paper's Definition 19 reads:

> *"Consider the graph $\mathfrak{G}$ over $[r]$ which has an edge $\{i,j\}$ whenever the ith and the jth arrows have the same source and let $n_1,\dots,n_r$ denote the rows of a matrix with columns the rays of the stoichiometric cone. If to the graph $\mathfrak{G}$ we adjoin the edges $\{i,s\}$ and $\{j,s\}$ whenever n_s is in the span of $\{n_i,n_j\}$ for each edge $\{i,j\}$ of $\mathfrak{G}$, we obtain the preclustering graph. A precluster is a connected component of the preclustering graph."*

Call this graph G^+. The thesis (Definition 4.3 and Algorithm 4.4 of [5]) iterates instead. Starting from $P = G$, while some edge $\{i,j\}$ of P and some s outside the component of i satisfy $n_s \in \operatorname{span}\{n_i,n_j\}$, it adjoins $\{i,s\}$ and $\{j,s\}$. Let $\overline{G}$ be the smallest graph that contains G and is closed under this adjunction rule without the component restriction.

Lemma 5. G^+, $\overline{G}$ and every output of Algorithm 4.4 (in any order of choices) have the same connected components. We write $\mathsf{P}(\mathcal{N})$ for this partition of the reactions.

Proof. For a doubling pair $e = \{i,j\}$ put $V_e = \operatorname{span}\{n_i,n_j\}$. We claim that every edge $\{a,b\}$ of $\overline{G}$ has $\operatorname{span}\{n_a,n_b\} \subseteq V_e$ for some doubling pair e with a,b in the G^+-component of e. This holds for the edges of G. Suppose it holds for an edge $\{a,b\}$ with pair $e = \{i,j\}$, and let $n_s \in \operatorname{span}\{n_a,n_b\} \subseteq V_e$. If $s \notin e$, then G^+ contains $\{i,s\}$, so in all cases s lies in the G^+-component of e. Also $\operatorname{span}\{n_a,n_s\}$ and $\operatorname{span}\{n_b,n_s\}$ lie in V_e. This proves the claim. So every edge of $\overline{G}$ lies inside a G^+-component. Since $E(G^+) \subseteq E(\overline{G})$, the two graphs have the same components.

Let A be an output of Algorithm 4.4. Then $E(A) \subseteq E(\overline{G})$, so the components of A refine those of $\overline{G}$. Conversely, every edge of G^+ lies inside a component of A. For edges of G this holds because $G \subseteq A$. For an adjoined edge $\{i,s\}$ with $n_s \in V_{\{i,j\}}$, the algorithm cannot stop while s lies outside the component of i. Hence the components of G^+ refine those of A, and all three partitions coincide. □

The enumeration of §4 confirms that the three constructions never differ. Under (H2) every reaction lies in an internal cycle, so $n_s \neq 0$ for all s. This is why the paper needs (H2): a reaction with $n_s = 0$ lies in every span and would be glued to every doubling pair.

Full coupling. Assume (H2). Reactions s,t are *fully coupled* if $n_s = \mu n_t$ for some $\mu > 0$. Equivalently, $v_s = \mu v_t$ for every $v \in \ker\Gamma$, since the extreme rays span $\ker\Gamma$ by (H2). This is the full coupling of flux coupling analysis [12], and the classes are the "enzyme subsets" of [13].

Proposition 6. If $\mathcal{N}$ satisfies (H2), then $\mathcal{N}^*$ satisfies (H1), and $\mathsf{M}(\mathcal{N}^*)$ is the partition of the reactions of $\mathcal{N}$ into classes of fully coupled reactions.

Proof. Apply Proposition 3(4) to $\mathcal{N}^*$. By Lemma 2, the extreme conservation laws of $\mathcal{N}^*$ are the extreme rays of $C(\mathcal{N})$, so the columns w_s^* are the rows n_s. □

In the same way, by Proposition 3(4), $\mathsf{M}(\mathcal{N})$ groups species i, j with $w_i = \mu w_j$. Call such species *conservation-coupled*.

2 Exact criteria

Theorem 7. Let $\mathcal{N}$ be consistent (H2). Let $\mathsf{P} = \mathsf{P}(\mathcal{N})$, and let M be the partition into fully coupled classes, which equals $\mathsf{M}(\mathcal{N}^*)$ by Proposition 6.

(1) $\mathsf{P} \preceq \mathsf{M}$ if and only if every doubling pair is fully coupled.

(2) $\mathsf{M} \preceq \mathsf{P}$ if and only if, for every fully coupled class κ with $|\kappa| \geq 2$, the common line $\mathbb{R}n_s$ ($s \in \kappa$) lies in $\operatorname{span}\{n_i, n_j\}$ for some doubling pair $\{i, j\}$. In particular $\mathsf{M} \preceq \mathsf{P}$ holds if every such class contains a reaction that shares its source with another reaction.

(3) $\mathsf{P} = \mathsf{M}$ if and only if every doubling pair is fully coupled and every fully coupled class with at least two elements contains a doubling pair.

Proof. By (H2), $n_s \neq 0$ for all s. Two nonzero nonnegative vectors that are linearly dependent are positive multiples of each other, so $n_s \in \mathbb{R}n_i$ means that s and i are fully coupled. We use G^+ (Lemma 5).

(1) If $\mathsf{P} \preceq \mathsf{M}$, the two reactions of a doubling pair lie in one precluster, hence in one coupled class. Conversely, assume that all doubling pairs are fully coupled. For a doubling pair $\{i, j\}$ we then have $\operatorname{span}\{n_i, n_j\} = \mathbb{R}n_i$. So every s adjoined through $\{i, j\}$ is fully coupled to i. Thus every edge of G^+ joins fully coupled reactions, and $\mathsf{P} \preceq \mathsf{M}$.

(2) Suppose the condition holds for a class κ with the pair $\{i, j\}$. Every $t \in \kappa$ lies in $\{i, j\}$ or is joined to i by an edge of G^+, so κ lies in one precluster. Conversely, suppose κ lies in one precluster and $|\kappa| \geq 2$. Then any $t \in \kappa$ is not isolated in G^+. By the definition of G^+, either t belongs to a doubling pair $\{t, j\}$, and then $n_t \in \operatorname{span}\{n_t, n_j\}$; or t was adjoined through a doubling pair $\{i, j\}$ with $n_t \in \operatorname{span}\{n_i, n_j\}$. In both cases the condition holds.

(3) Assume both conditions. Then $\mathsf{P} \preceq \mathsf{M}$ by (1). If a class κ with $|\kappa| \geq 2$ contains the doubling pair $\{i, j\}$, then $n_\kappa \in \operatorname{span}\{n_i, n_j\}$, so (2) gives $\mathsf{M} \preceq \mathsf{P}$. Conversely, if $\mathsf{P} = \mathsf{M}$, then (1) gives the first condition. Take a class κ with $|\kappa| \geq 2$ and $t \in \kappa$. As in (2), t is in a doubling pair or adjoined through one, $\{i, j\}$, with $n_t \in \operatorname{span}\{n_i, n_j\} = \mathbb{R}n_i$. So t is fully coupled to i and to j, and $\{i, j\} \subseteq \kappa$. □

Two species are *twins* if they have identical rows of Y_e, that is, equal coefficients in every source complex. The twin pairs of $\mathcal{N}$ are exactly the doubling pairs of $\mathcal{N}^*$.

Corollary 8 (species side)**.** Let $\mathcal{N}$ be conservative (H1). Then:

(1) $\mathsf{P}(\mathcal{N}^*) \preceq \mathsf{M}(\mathcal{N})$ if and only if all twin pairs are conservation-coupled;

(2) $\mathsf{M}(\mathcal{N}) \preceq \mathsf{P}(\mathcal{N}^*)$ if and only if, for every MIPS Σ with $|\Sigma| \geq 2$, the line $\mathbb{R}w_\Sigma$ lies in $\operatorname{span}\{w_i, w_j\}$ for some twin pair $\{i, j\}$;

(3) $\mathsf{P}(\mathcal{N}^*) = \mathsf{M}(\mathcal{N})$ if and only if all twin pairs are conservation-coupled and every MIPS with at least two elements contains a twin pair.

Proof. Apply Theorem 7 to $\mathcal{N}^*$, which is consistent by Lemma 2. Its doubling pairs are the twin pairs. Its vectors n_i^* are the columns w_i. Its fully coupled classes are $\mathsf{M}(\mathcal{N})$, by Proposition 6 applied to $\mathcal{N}^*$ together with $\mathcal{N}^{**} = \mathcal{N}$. □

Corollary 9 (product complexes are sources)**.** Assume (H1) and (H3), and assume that every product complex of $\mathcal{N}$ is also a source complex. This holds, for example, when $\mathcal{N}$ is weakly reversible. Then $\mathcal{N}$ has no twins, so $\mathsf{P}(\mathcal{N}^*)$ is the partition into singletons. Hence $\mathsf{P}(\mathcal{N}^*) \preceq \mathsf{M}(\mathcal{N})$ always holds. Moreover, $\mathsf{P}(\mathcal{N}^*) = \mathsf{M}(\mathcal{N})$ holds if and only if $\operatorname{im}\Gamma$ contains no nonzero vector supported on exactly two coordinates.

Proof. Suppose X_i, X_j are twins. Every product complex is a source complex, so its ith and jth coordinates are equal. Hence rows i and j of Y_{p} are also equal, and so are rows i and j of Γ. This contradicts (H3). The last claim follows from Proposition 3(3). Under (H1), a nonzero vector of $\operatorname{im}\Gamma$ cannot be supported on a single coordinate, since $z^{\mathsf{T}} e_i = z_i > 0$ for the positive conservation law z. A vector supported on $\{i, j\}$ has the form $a(e_i - \mu e_j)$ with $\mu > 0$. □

Corollary 10 (weakly reversible networks, reaction side)**.** Let $\mathcal{N}$ be weakly reversible, with $y_j \neq y'_j$ for all j.

(1) If some complex is the source of two reactions, then $\mathsf{P}(\mathcal{N}) \not\preceq \mathsf{M}(\mathcal{N}^*)$.

(2) If moreover $\mathcal{N}$ has deficiency zero, then $\mathsf{P}(\mathcal{N}) \neq \mathsf{M}(\mathcal{N}^*)$ in all cases.

Proof. For a directed cycle γ in the reaction graph, the indicator vector 1_γ lies in $C(\mathcal{N})$, because the reaction vectors $y' - y$ along γ sum to zero. A weakly reversible network is consistent, since the sum of such indicators over a cycle cover is positive. (1) Let $R_a : y \to y_1$ and $R_b : y \to y_2$. By weak reversibility there is a shortest directed path from y_1 to y. It does not pass through y before its end, so it does not use R_b. Together with R_a it forms a cycle γ with $(1_\gamma)_a = 1$ and $(1_\gamma)_b = 0$. So R_a, R_b are not fully coupled, and Theorem 7(1) applies. (2) If some complex has two outgoing reactions, use (1). Otherwise every complex has exactly one outgoing reaction, since every complex of a weakly reversible network is a source. Each linkage class is strongly connected, so it is a directed cycle of length ≥ 2. With deficiency zero, $\ker\Gamma$ is the kernel of the incidence map [14]. So every $v \in \ker\Gamma$ has equal inflow and outflow at each complex, and all reactions of one cycle are fully coupled. This gives a class with at least two elements and no doubling pair, so Theorem 7(3) fails. □

3 Counterexamples

The networks of Table 11 are the following. Species are $X_1, X_2, \ldots$, and reactions are numbered in the order listed.

- $R : X_1 \to X_2,\ X_2 \to X_1$.

	Formulation	Fails for
Conjecture 22, species side (ground set: species of $\mathcal{N}$)		
A1	$\mathsf{P}(\mathcal{N}^*) = \mathsf{M}(\mathcal{N})$	R, T, U; F (ray, semiflow)
A2	$\mathsf{P}(\mathcal{N}^*) \preceq \mathsf{M}(\mathcal{N})$	W^*, U; F (ray, semiflow)
A3	$\mathsf{M}(\mathcal{N}) \preceq \mathsf{P}(\mathcal{N}^*)$	R, T
A4	$\lvert\mathsf{P}(\mathcal{N}^*)\rvert = \lvert\mathsf{M}(\mathcal{N})\rvert$	R, T
Conjecture 22, reaction side (ground set: reactions of $\mathcal{N}$)		
B1	$\mathsf{P}(\mathcal{N}) = \mathsf{M}(\mathcal{N}^*)$	R, T, W, N_3; F (ray, semiflow); every WR deficiency-0 network (ray)
B2	$\mathsf{P}(\mathcal{N}) \preceq \mathsf{M}(\mathcal{N}^*)$	W, N_3; F (ray, semiflow)
B3	$\mathsf{M}(\mathcal{N}^*) \preceq \mathsf{P}(\mathcal{N})$	R, T, $N_{4.12}$
B4	$\lvert\mathsf{P}(\mathcal{N})\rvert = \lvert\mathsf{M}(\mathcal{N}^*)\rvert$	R, T
C	$\lvert\mathsf{P}(\mathcal{N})\rvert = \lvert\mathsf{M}(\mathcal{N})\rvert$ (no dualization)	R, T
Question 24, both sides: $\mathsf{P}(\mathcal{N}^*)$ vs $\mathsf{S}(\mathcal{N})$, and $\mathsf{P}(\mathcal{N})$ vs $\mathsf{S}(\mathcal{N}^*)$		
Q1	$\mathsf{P} = \mathsf{S}$	T
Q2	every minimal siphon lies in a precluster ($\mathsf{S} \preceq \mathsf{P}$)	T
Q3	every precluster lies in a minimal siphon ($\mathsf{P} \preceq \mathsf{S}$)	D, F
Q4	every precluster is a siphon	T
Q5	every precluster is a union of minimal siphons	T
Q6	$\lvert\mathsf{P}\rvert = \lvert\mathsf{S}\rvert$	T
Remark 23		
R23	every internal cycle of $\mathcal{N}$ lies in a precluster of $\mathcal{N}$	T, $N_{4.12}$, R

Table 11 / Precise formulations of Conjecture 22, Question 24 and Remark 23, and counterexamples satisfying (H1)–(H3). P denotes preclusters (any of the three constructions, Lemma 5). M denotes MIPS in any of the three readings of Remark 4, except where a reading is indicated. S denotes the minimal siphons. The networks are defined in the text. T, W, U and D have pairwise linearly independent rows of Γ, and so do the duals of T, U and D.

- $T\colon X_1 \to X_2,\ X_2 \to X_3,\ X_3 \to X_1$, the 3-cycle.
- $W\colon X_1 \to X_2,\ X_1 \to X_3,\ X_2 \to X_1,\ X_3 \to X_1$, and its dual $W^*\colon K_1 + K_2 \to K_3 + K_4,\ K_3 \to K_1,\ K_4 \to K_2$.
- $U\colon X_4 \to X_2 + X_3,\ X_3 + X_4 \to X_1,\ X_1 + X_2 \to 2X_4$.
- $D\colon X_1 \to X_2,\ X_1 \to X_3,\ X_2 + X_3 \to 2X_1$.
- F, the one-site futile cycle: $S_0 + E \rightleftarrows ES_0 \to S_1 + E,\ S_1 + F \rightleftarrows FS_1 \to S_0 + F$.
- $N_3\colon 2X_1 \rightleftarrows X_1 + X_2 \to 2X_2$, the running example of the thesis [5] (Examples 1.22, 4.1, 4.5, 4.7).
- $N_{4.12}\colon 2X_1 \rightleftarrows X_1 + X_2$ (thesis Example 4.12).

Proof (Proof of Theorem 1)**.** Each entry of Table 11 is checked by hand below and by exact computation (`certify_examples.py`, 142 claims). For every network, $k = \sum_{\text{extreme flux rays}} v$ and $x = (1, \dots, 1)$ give a positive steady state, so (H2) holds. Conservativity and (H3) are checked directly.

T. Γ has rows $(-1, 0, 1), (1, -1, 0), (0, 1, -1)$. They are pairwise linearly independent, and so are the columns. $\ker \Gamma$ and the conservation space are both spanned by $(1, 1, 1)$.

T is weakly reversible with deficiency $3-1-2=0$, so it is complex balanced and has positive steady states for every k [15] [14]. The three sources are distinct, so G has no edge and $\mathsf{P}(T)$ consists of singletons. The flux cone is the ray $(1,1,1)$, so all n_s are equal and $\mathsf{M}(T^*)$ is one block. The dual is $T^*\colon K_1 \to K_3 \to K_2 \to K_1$, again a 3-cycle. So the species side is the same: $\mathsf{P}(T^*)$ consists of singletons, while $\mathsf{M}(T) = \{X_1, X_2, X_3\}$. The latter holds in all three readings, because $q = 1$ and the only extreme conservation law is $(1,1,1)$. A nonempty siphon containing X_i must contain the source of the reaction producing X_i, so $\{X_1, X_2, X_3\}$ is the only siphon of T, and likewise of T^*. This refutes A1, A3, A4, B1, B3, B4, C, Q1, Q2, Q4, Q5 and Q6. The internal cycle $\{R_1, R_2, R_3\}$ lies in no precluster, which refutes R23.

R. The same argument with two species: $\mathsf{P}(R) = \{\{R_1\}, \{R_2\}\}$, $\mathsf{M}(R^*) = \{\{R_1, R_2\}\}$, and $R^* \cong R$. Among networks with nonzero reaction vectors, no network with fewer species or reactions satisfies (H1) and (H2): with one species, (H1) forces $\Gamma = 0$, and with one reaction, (H2) forces its reaction vector to vanish. Zero-vector loops, allowed in the general definitions above, are excluded from this minimality claim.

W and W.* The flux cone of W has extreme rays $(1,0,1,0)$ and $(0,1,0,1)$. So $n_1 = n_3 = (1,0)$ and $n_2 = n_4 = (0,1)$. The doubling pair $\{R_1, R_2\}$ is not fully coupled, so $\mathsf{P}(W) \not\preceq \mathsf{M}(W^*)$ by Theorem 7(1). Explicitly, $\mathsf{P}(W)$ is one block, while $\mathsf{M}(W^*) = \{\{R_1, R_3\}, \{R_2, R_4\}\}$ in all three readings. W is weakly reversible with deficiency zero, and its rows of Γ are pairwise linearly independent and its columns distinct. By Lemma 2, W^* satisfies (H1)–(H3). Its species side is the reaction side of W, which refutes A2.

U. The flux cone is the ray $(1,1,1)$. The extreme conservation laws are $(1,1,0,1)$ and $(2,0,1,1)$, so $w_1 = (1,2)$, $w_2 = (1,0)$, $w_3 = (0,1)$ and $w_4 = (1,1)$. These are pairwise non-proportional, so $\mathsf{M}(U)$ consists of singletons (the set reading agrees). X_1 and X_2 are twins (both occur only in the source $X_1 + X_2$) and are not conservation-coupled. So $\mathsf{P}(U^*) \not\preceq \mathsf{M}(U)$ by Corollary 8(1). Indeed $\operatorname{span}\{w_1, w_2\} = \mathbb{R}^2$, so $\mathsf{P}(U^*)$ is one block. In the semiflow reading the supports are $\{X_1, X_2, X_4\}$ and $\{X_1, X_3, X_4\}$, and neither contains all four species. The rows and the columns of $\Gamma(U)$ are pairwise linearly independent, so U and U^* satisfy the strongest form of (H3). By Corollary 9, a counterexample to A2 cannot be weakly reversible.

D. The flux cone is the ray $(1,1,1)$ and the only conservation law is $X_1 + X_2 + X_3$. So the doubling pair $\{R_1, R_2\}$ and the twins X_2, X_3 are coupled, and $\mathsf{P}(D) = \mathsf{M}(D^*)$ and $\mathsf{P}(D^*) = \mathsf{M}(D)$ are the one-block partitions (Theorem 7(3)). The two equalities of Conjecture 22 hold for D. The dual $D^*\colon K_1 + K_2 \to 2K_3$, $K_3 \to K_1$, $K_3 \to K_2$ is isomorphic to D. The minimal siphons of D are $\{X_1, X_2\}$ and $\{X_1, X_3\}$: $\{X_1\}$ is not a siphon because X_1 is produced from $X_2 + X_3$, and $\{X_2\}$, $\{X_3\}$ are not siphons because they are produced from X_1. The single precluster lies in neither, which refutes Q3 on both sides. The rows and the columns of $\Gamma(D)$ are pairwise linearly independent.

F. Order the species $S_0, S_1, E, F, ES_0, FS_1$ and the reactions as written. The extreme flux rays are $R_1 + R_2$, $R_4 + R_5$ and $R_1 + R_3 + R_4 + R_6$. The extreme conservation laws are $E + ES_0$, $F + FS_1$ and $S_0 + S_1 + ES_0 + FS_1$. The doubling pairs are $\{R_2, R_3\}$ and $\{R_5, R_6\}$, and neither is fully coupled. The only fully coupled pair is $\{R_3, R_6\}$. So $\mathsf{P}(F)$ is one block, while $\mathsf{M}(F^*) = \{\{R_3, R_6\}, \{R_1\}, \{R_2\}, \{R_4\}, \{R_5\}\}$. The twins are (S_0, E) and (S_1, F), and they are not conservation-coupled. So $\mathsf{P}(F^*)$ is one block, while

$\mathsf{M}(F) = \{\{S_0, S_1\}, \{E\}, \{F\}, \{ES_0\}, \{FS_1\}\}$. In the semiflow reading the supports are those of the three conservation laws, and none contains all species. Likewise, on the reaction side no internal cycle of F contains all reactions. The minimal siphons of F are $\{E, ES_0\}$, $\{F, FS_1\}$ and $\{S_0, S_1, ES_0, FS_1\}$. Those of F^* are $\{R_1, R_2\}$, $\{R_4, R_5\}$ and $\{R_1, R_3, R_4, R_6\}$. In both cases the single precluster lies in no minimal siphon, which refutes Q3.

N_3 and $N_{4.12}$. For N_3 the internal cycles are $\{R_1, R_2\}$ and $\{R_2, R_3\}$ (thesis Example 1.22). So $n_1 = (1, 0)$, $n_2 = (1, 1)$, $n_3 = (0, 1)$, and the doubling pair $\{R_2, R_3\}$ is not fully coupled. Hence $\mathsf{P}(N_3) = \{\{R_1, R_2, R_3\}\}$ (thesis Example 4.5), while $\mathsf{M}(N_3^*)$ consists of singletons (also in the set reading; in the semiflow reading it consists of the two internal cycles, neither of which contains all three reactions). For $N_{4.12}$ the thesis finds the preclusters $\{R_1\}, \{R_2\}$. The flux cone is the ray $(1, 1)$, so $\mathsf{M}(N_{4.12}^*) = \{\{R_1, R_2\}\}$, and the internal cycle $\{R_1, R_2\}$ lies in no precluster, although $N_{4.12}$ has positive steady states for all k. Both networks satisfy (H1)–(H3).

The entry "every WR deficiency-0 network" for B1 is Corollary 10(2). The entries U for A1 and W, N_3 for B1 follow from the failures of A2 and B2, since equality implies refinement. □

Remark 12. In Example 25 of [1], $\mathcal{N}^*$ has the single precluster $\{x_1, x_2, x_3, x_4\}$, while the MIPS of $\mathcal{N}$ are $\{x_1, x_3\}$ and $\{x_2, x_4\}$. This already contradicts A1 and A2, but that network has no positive steady state ($\dot{x}_2 = k_1 x_3 x_4$), so it lies outside Conjecture 22. The counterexamples above lie inside. The diagram at the end of §3 of [1] also links "internal cycles" and "MIPS" by an arrow labelled "dual". For F, the internal cycles of F^* are the supports of the conservation laws above, which differ from the ray-reading MIPS of F. ⌟

Remark 13 (clusters)**.** Conradi and Flockerzi [6] (§3.2, Assumption 2, Fact 2, Algorithm 1) call $e_l - e_k \in \ker Y_e$ a doubling vector; these are our doubling pairs. Their Fact 2(a) is the span rule of Definition 19: on the pairs $(\lambda, \mu) \in \Lambda(E)^2$ with $n_l\lambda / n_l\mu = n_k\lambda / n_k\mu$, every j with $n_j \in \operatorname{span}\{n_l, n_k\}$ has the same ratio. Their clusters group the rows with equal ratio on the pairs that satisfy these doubling relations, where $\Lambda(E) = \{\lambda \geq 0 : E\lambda > 0\}$. This definition does not involve rate constants. The thesis (Definition 4.6 [5]) fixes k instead (its Remark 4.9). In both versions every precluster lies in a cluster. For the two-site phosphorylation network of Table 14 (their network N_1, with the same reaction order), their clusters $\{1, 2, 3, 10, 11, 12\}$ and $\{4, \ldots, 9\}$ (their Example 1) coincide with our preclusters. The 2024 talk [3] conjectures that siphons and clusters are dual. The cluster version of B1 fails as well. For N_3, the doubling relation $(\lambda_1 + \lambda_2)/(\mu_1 + \mu_2) = \lambda_2/\mu_2$ with $\lambda, \mu > 0$ forces $\mu \in \mathbb{R}_{>0}\lambda$, so all three ratios coincide and the Conradi–Flockerzi cluster is $\{R_1, R_2, R_3\}$. The thesis finds the same single cluster for $k_2 > k_3$ (Example 4.7). Yet no two reactions of N_3 are fully coupled. ⌟

Table 14 shows failures on standard motifs. The preclusters are strictly coarser than the MIPS on one side of the duality and strictly finer on the other, or strictly coarser on both sides (the futile cycle). Equality occurred only for network G, where both partitions consist of singletons.

4 Exhaustive check on small networks

`enumerate_small.py` lists every network, up to permutation of species, with distinct

Network	reaction side: $\mathsf{P}(\mathcal{N})$ vs $\mathsf{M}(\mathcal{N}^*)$	species side: $\mathsf{P}(\mathcal{N}^*)$ vs $\mathsf{M}(\mathcal{N})$
$E+S \rightleftarrows ES \rightleftarrows E+P$	$\mathsf{M} \prec \mathsf{P} = \{\text{all}\}$	$\mathsf{P} = \text{singletons} \prec \mathsf{M} \ni \{S, P\}$
one-site futile cycle F	$\mathsf{M} \prec \mathsf{P} = \{\text{all}\}$	$\mathsf{M} \prec \mathsf{P} = \{\text{all}\}$
two-site distributive phosphorylation (9 species, 12 reactions)	$\mathsf{M} \prec \mathsf{P}$: 2 preclusters (the E- and F-steps $S_0 \leftrightarrow S_1$ together, and $S_1 \leftrightarrow S_2$ together); coupled pairs $\{R_3, R_{12}\}, \{R_6, R_9\}$	$\mathsf{P} = \text{singletons} \prec \mathsf{M}$; MIPS $\{S_0, S_1, S_2\}$, $\{ES_0, ES_1\}$, $\{FS_2, FS_1\}$, $\{E\}$, $\{F\}$
thesis network G: $3X_1 \rightleftarrows 2X_1 + X_2$, $X_1 + X_2 \to 2X_2$, $X_2 \to X_1$	$\mathsf{P} = \mathsf{M} = \text{singletons}$	$\mathsf{P} = \text{singletons} \prec \mathsf{M} = \{\{X_1, X_2\}\}$
$2A \to 2B \to A + C \to 2A$	$\mathsf{P} = \text{singletons} \prec \mathsf{M}$	$\mathsf{P} = \text{singletons} \prec \mathsf{M}$

Table 14 / Standard motifs (all satisfy (H1)–(H3); `motifs.py`). "$\mathsf{P} \prec \mathsf{M}$" means strict refinement.

non-loop reactions between nonzero complexes. Conservativity excludes a non-loop reaction incident to the zero complex. Every species must occur. The families are: two species, molecularity ≤ 2, $r \leq 5$; three species, molecularity ≤ 2, $r \leq 4$; three species, molecularity ≤ 3, $r \leq 3$; four species, molecularity ≤ 1, all r; four species, molecularity ≤ 2, $r \leq 3$. For the 15 513 networks that satisfy (H1)–(H3), the script computes both sides of the duality, the three precluster constructions, the three MIPS readings (the set reading when $q \leq 2$), the minimal siphons and the internal cycles, all in exact rational arithmetic.

The results are as follows. The three precluster constructions never differ (Lemma 5). The criteria of Theorem 7 and Corollary 8 agree with the direct computation on every network, on both sides and for every construction. Every formulation of Table 11 fails for some network. Table 15 gives the counts.

Formulation	all	strong	WR	WR0
networks satisfying (H1)–(H3)	15 513	11 121	339	304
A1 $\mathsf{P}(\mathcal{N}^*) = \mathsf{M}(\mathcal{N})$	428	338	3	3
A2 $\mathsf{P}(\mathcal{N}^*) \preceq \mathsf{M}(\mathcal{N})$	15 380	11 118	339	304
A3 $\mathsf{M}(\mathcal{N}) \preceq \mathsf{P}(\mathcal{N}^*)$	561	341	3	3
B1 $\mathsf{P}(\mathcal{N}) = \mathsf{M}(\mathcal{N}^*)$	2 532	1 278	10	0
B2 $\mathsf{P}(\mathcal{N}) \preceq \mathsf{M}(\mathcal{N}^*)$	13 833	10 757	201	174
B3 $\mathsf{M}(\mathcal{N}^*) \preceq \mathsf{P}(\mathcal{N})$	4 211	1 642	147	129
Q1 $\mathsf{P}(\mathcal{N}^*) = \mathsf{S}(\mathcal{N})$	126	97	0	0
Q3 $\mathsf{P}(\mathcal{N}^*) \preceq \mathsf{S}(\mathcal{N})$	7 809	5 517	229	204
R23 internal cycles $\preceq \mathsf{P}(\mathcal{N})$	2 346	1 157	59	59

Table 15 / Number of networks (out of those satisfying (H1)–(H3)) for which a formulation holds, ray reading, summed over the five families (`certificates/enumeration_summary.log`). "Strong": rows and columns of Γ pairwise linearly independent. "WR": weakly reversible. "WR0": weakly reversible with deficiency zero.

The pattern matches the corollaries. A2 never fails for weakly reversible networks (Corollary 9). B1 never holds for weakly reversible deficiency-zero networks (Corollary 10). A1 holds for a weakly reversible network exactly when $\mathsf{M}(\mathcal{N})$ consists of singletons.

5 What is settled and what remains open

Settled. The conjecture is false under every reading of "duality relation" as an equality, a refinement in either direction, or an equality of cardinalities between the precluster partition and the MIPS partition, on either side of the network duality and for every reading of MIPS. This includes the reading by analogy with Theorem 21, $\mathsf{P}(\mathcal{N}^*) = \mathsf{M}(\mathcal{N})$ or $\mathsf{P}(\mathcal{N}) = \mathsf{M}(\mathcal{N}^*)$, which we consider the most plausible. Theorem 7 and Corollary 8 determine exactly the networks for which these relations hold. The six siphon relations of Question 24 fail as well, and so does the claim of Remark 23.

Open: the relation itself. Conjecture 22 does not define "duality relation", and the closing diagram of the source writes "Refine" both literally (internal cycles refine preclusters) and in quotation marks (between MIPS and siphons). Our results exclude every relation that forces a refinement or a bijection between blocks. They do not exclude a relation of another kind, for instance a Galois correspondence that is not a refinement, or a refinement that holds outside an explicit list of degenerate networks. Excluding small or degenerate networks does not by itself rescue the precise forms: among the 11 121 enumerated networks whose rows and columns of Γ are pairwise linearly independent, the species-side equality A1 holds for 338 and the reaction-side equality B1 for 1 278 (Table 15). We examined clusters (in the thesis and in Conradi–Flockerzi) only on N_3 and on the two-site phosphorylation network (Remark 13).

Open: the Global Attractor Conjecture. The source motivates the conjecture by a possible new case of the Global Attractor Conjecture (GAC) in small codimension, by analogy with the known cases in small dimension (§3 of [1]). We have no result on this. Three elementary facts constrain an argument through the network duality $\mathcal{N} \mapsto \mathcal{N}^*$. The first two are checked on the networks of this paper by `certify_examples.py`; the third restates two corollaries of §2.

(1) The duality preserves the dimension. Since $\Gamma(\mathcal{N}^*) = \Gamma^{\mathsf{T}}$, a network with n species, r reactions and a stoichiometric subspace of dimension s is sent to a network with r species, n reactions and the same dimension s. The case of dimension at most three [16] is therefore mapped to itself. The codimension $n - s$ of $\mathcal{N}$ becomes the dimension of the cycle space $\ker \Gamma(\mathcal{N}^*)$, not a dimension of $\mathcal{N}^*$.

(2) The duality does not preserve weak reversibility, which complex balancing requires [15]. Blanchini and Giordano already observe that the dual of a reversible network is not even weakly reversible, and that mass-action rate functions are not preserved (p. 6 [8]). The network W is weakly reversible with deficiency zero, hence complex balanced for all rate constants, whereas $W^*\colon R_1 + R_2 \to R_3 + R_4,\ R_3 \to R_1,\ R_4 \to R_2$ is not weakly reversible, hence complex balanced for no rate constants. The 3-cycle T is self-dual, so weak reversibility is sometimes preserved.

(3) For weakly reversible $\mathcal{N}$, the preclusters of $\mathcal{N}^*$ are singletons (Corollary 9). At deficiency zero, moreover, $\mathsf{P}(\mathcal{N}) \neq \mathsf{M}(\mathcal{N}^*)$ (Corollary 10). In the setting of the GAC, the precluster partition carries no information on the species side.

A transfer of known cases of the GAC through duality would therefore have to act on other objects than the partitions compared here. Two candidates are Lyapunov functions, which Blanchini and Giordano relate between a network and its dual [8],

and the siphon conditions used in recent special cases of the GAC [17]. The GAC itself is now claimed in full by two preprints: a major revision of Craciun's proof [18], and a proof of the boundedness and persistence conjectures for weakly reversible mass-action systems [19], from which the GAC follows by the classical argument. A new special case is due to Wiuf [17]. We have not checked these preprints. If the GAC is settled, the question raised by the source becomes which structure the duality transfers, rather than which new cases of the GAC it yields.

Limitations. Our reading of Definition 17 (Proposition 3) quantifies over all $c \in Q$, including values on the boundary of chambers. This is the reading under which Remark 18 holds. The counterexamples were checked under the other two readings as well. The paper defines preclusters only through Definition 19. We used the thesis for the iterated version and for the examples N_3 and $N_{4.12}$. The original definitions of Conradi and Flockerzi [6] (doubling vectors, the span rule, clusters) agree with those of the thesis on the points used here (Remark 13). The criteria, the counterexamples and the enumeration were checked by a second agent with a fresh context on 25 October 2026 (`verification.md`). The remarks on the GAC in §5 were added on 27 October 2026, after that check.

6 Computer verification

All scripts are in `scripts/`. They use only the Python standard library and exact rational arithmetic (`fractions.Fraction`), print PASS/FAIL per claim, and exit with a nonzero status on failure. Logs are in `certificates/`. They were run under Python 3.12 on Fedora 44 (WSL).

- `crn.py`: the library. It computes extreme rays of the flux and conservation cones by minimal supports (a support S is extreme if and only if the kernel restricted to S is one-dimensional and spanned by a vector of constant sign with support S), the dual network, siphons by exhaustive search, the three precluster constructions, the three MIPS readings (the set reading via exact vertex supports of P_c for representatives of every cell of the chamber decomposition when $q \leq 2$), and the hypotheses.
- `certify_examples.py`: 142 claims. These include consistency with Examples 15 and 25 of the paper and Examples 1.22, 4.1, 4.5 and 4.12 of the thesis, every entry of Table 11, and an explicit positive steady state for each counterexample. Under 1 s.
- `motifs.py`: Table 14. Under 10 s.
- `enumerate_small.py`: the enumeration of §4, with the check of Theorem 7 and Corollary 8 (PASS on all 15 513 networks). It takes 5 s to 75 s per family on 12 cores.
- `summarize_enumeration.py`: Table 15, and the check that every formulation fails at least once.

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